Table 1. Comparative experiment results.

Trace	Attack	No defense	FTMaster	FTOP	FTODefender	FloRa	DMS-BTT	rDefender	FEVER
Benign flows harmed (%)									
T1B	G2	10.60	0.02	2.11	0.26	0.20	0.60	4.02	0.00
	Mimic	5.60	2.34	4.63	2.80	4.15	1.54	3.04	0.02
	Spoof	12.64	0.90	0.70	1.02	2.40	0.10	4.03	0.00
	Slow	0.00	0.04	0.86	0.31	0.15	0.09	0.00	0.01
	Pulse	3.89	0.03	2.02	0.23	3.18	0.02	1.64	0.01
	Aware	12.91	7.88	1.73	9.09	12.08	1.87	2.27	0.01
	EVX	0.00	0.01	0.00	0.25	0.09	0.02	0.00	0.01
UHt	G2	38.86	0.00	--	--	--	0.00	14.87	0.00
	Mimic	29.60	25.19	--	--	--	9.24	10.80	0.10
	Aware	57.15	46.30	--	--	--	9.22	9.96	0.01
Attack occupancy (%)									
T1B	G2	39.5	24.9	30.8	0.3	14.0	26.8	34.1	4.8
	Mimic	37.7	36.4	32.7	36.1	36.9	33.3	34.1	4.9
	Spoof	39.7	28.6	29.9	11.2	21.4	25.9	33.8	4.6
	Slow	21.2	20.8	21.0	20.5	20.8	20.3	21.1	19.7
	Pulse	30.3	18.4	24.1	0.2	29.7	20.1	28.7	1.9
	Aware	49.9	48.3	44.3	48.8	49.0	43.8	42.8	14.1
	EVX	20.6	20.5	20.6	0.2	14.3	14.1	20.6	17.7
UHt	G2	44.5	27.1	--	--	--	30.7	37.5	7.7
	Mimic	42.1	41.4	--	--	--	36.7	36.8	8.4
	Aware	53.9	53.3	--	--	--	43.4	43.2	7.0
New-connection success (%)									
T1B	G2	31.1	41.2	31.3	100.0	70.0	28.6	31.4	98.8
	Mimic	32.3	31.6	32.3	29.1	30.7	27.8	32.9	98.0
	Spoof	31.5	29.2	31.7	20.5	20.4	28.8	31.2	97.0
	Slow	70.6	69.0	71.3	61.2	68.3	61.9	70.0	74.3
	Pulse	12.7	61.6	11.6	100.0	12.0	11.2	12.1	96.9
	Aware	12.8	12.1	12.8	12.0	12.4	10.9	12.5	46.4
	EVX	96.5	88.1	81.8	100.0	93.3	95.1	99.7	98.6
UHt	G2	60.4	98.3	--	--	--	75.1	51.8	99.9
	Mimic	61.0	53.4	--	--	--	48.9	55.9	99.7
	Aware	20.2	19.0	--	--	--	17.9	19.3	89.8

Table 2. Controller overhead.

Method	Decision (ms)	CPU (%)	Memory (MB)	Packet-In (thousands)
No defense	19.4 (5.4)	76.5 (12.1)	199 (30)	99.8 (16.5)
FTMaster	53.7 (46.3)	78.3 (11.7)	205 (33)	96.0 (17.3)
FTOP	103.3 (81.4)	77.8 (12.3)	233 (41)	98.4 (16.8)
FTODefender	55.8 (46.7)	70.0 (16.6)	191 (34)	84.0 (13.5)
FloRa	35.2 (17.4)	75.3 (13.3)	200 (37)	94.7 (15.4)
DMS-BTT	58.3 (41.8)	77.8 (11.5)	232 (52)	108.3 (28.6)
rDefender	50.1 (38.9)	77.1 (11.7)	214 (50)	103.1 (18.8)
FEVER	20.8 (15.5)	62.2 (10.6)	253 (19)	87.7 (13.7)

Table 3. Federation cost of the detector.

Training	Shared	Rounds	Data sent (MB)	Time (s)	F1 (%)	Recall, other controllers (%)	Recall, Mimic (%)
Local only	nothing	--	0	--	78.4	75.8	0.2
FedAvg	weights	30	0.4	32	74.7	70.7	0.1
FedProx	weights	30	0.4	27	75.9	71.4	1.1
Federated XGBoost bagging	trees	100	173	212	78.5	75.8	0.1
Lossless federated XGBoost (FEVER)	histograms	300	2,601	38	86.8	90.5	60.9
Pooled XGBoost	rows and labels	1	27.6	--	86.5	90.5	61.1

Table 4. Ablation study.

Attack	Variant	Harm (%)	Occupancy (%)	New-connection success (%)	Lost first packets (%)	First block (s)
G2	FEVER	0.003 (0.006)	4.82 (0.95)	98.8 (1.1)	0.15 (0.16)	79 (9)
	w/o sharing	0.017 (0.015)	4.61 (1.44)	98.0 (2.0)	0.15 (0.18)	77 (13)
	w/o volume rule	0.003 (0.006)	4.80 (1.05)	98.1 (2.0)	0.12 (0.09)	78 (10)
	w/o detector	0.000 (0.000)	10.88 (1.36)	79.9 (2.1)	1.12 (1.47)	116 (6)
	w/o ingress blocking	0.003 (0.006)	24.79 (0.16)	84.1 (23.0)	1.50 (1.60)	77 (11)
Mimic	FEVER	0.020 (0.016)	4.90 (0.51)	98.0 (1.9)	0.03 (0.04)	80 (5)
	w/o sharing	3.090 (5.214)	15.67 (1.62)	63.4 (5.9)	4.77 (1.71)	144 (12)
	w/o volume rule	0.012 (0.008)	4.79 (0.74)	98.3 (2.0)	0.03 (0.05)	80 (6)
	w/o detector	0.000 (0.000)	14.12 (1.53)	67.7 (4.9)	2.63 (1.11)	135 (8)
	w/o ingress blocking	0.012 (0.022)	22.90 (2.43)	62.0 (23.7)	1.43 (1.63)	79 (7)
Aware	FEVER	0.010 (0.017)	14.12 (3.12)	46.4 (10.4)	15.68 (5.43)	91 (14)
	w/o sharing	2.744 (4.212)	18.18 (8.96)	35.2 (21.0)	25.36 (17.91)	92 (17), 4 of 5
	w/o volume rule	1.452 (1.778)	42.64 (5.08)	11.3 (2.1)	50.31 (3.94)	205 (33), 4 of 5
	w/o detector	0.000 (0.000)	10.40 (2.29)	55.4 (15.0)	9.76 (5.83)	76 (10)
	w/o ingress blocking	0.020 (0.028)	34.70 (1.91)	11.2 (2.0)	34.47 (8.68)	97 (13)

Table 5. Detection performance on T1B.

Method	Training	Threshold	Test attacks	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	False alarms (%)
FTMaster	C3 only	0.5 (paper)	All	75.32	99.80	64.66	78.48	0.33
FTOP	C3 only	0.5 (paper)	All	75.35	99.84	64.67	78.49	0.32
FloRa	C3 only	0.5 (paper)	All	75.34	99.82	64.68	78.50	0.32
DMS-BTT	C3 only	0.5 (paper)	All	75.35	99.82	64.68	78.50	0.32
FTMaster	C1–C3 pooled	0.5 (paper)	All	89.40	98.36	86.20	91.88	3.09
FTOP	C1–C3 pooled	0.5 (paper)	All	88.89	98.16	85.63	91.47	3.65
FloRa	C1–C3 pooled	0.5 (paper)	All	87.50	98.75	83.09	90.25	2.58
DMS-BTT	C1–C3 pooled	0.5 (paper)	All	87.85	98.31	83.98	90.58	3.25
FTMaster	C1–C3 pooled	0.1 % benign	All	80.95	99.83	72.74	84.16	0.22
FTOP	C1–C3 pooled	0.1 % benign	All	–	–	–	–	–
FloRa	C1–C3 pooled	0.1 % benign	All	79.26	99.87	70.28	82.51	0.20
DMS-BTT	C1–C3 pooled	0.1 % benign	All	78.10	99.81	68.65	81.35	0.30
FEVER	C1–C3 federated	0.1 % benign	All	83.73	99.74	76.82	86.79	0.79
FTMaster	C3 only	0.5 (paper)	G3, Pulse	99.81	99.82	99.86	99.84	0.33
FTOP	C3 only	0.5 (paper)	G3, Pulse	99.82	99.85	99.85	99.85	0.32
FloRa	C3 only	0.5 (paper)	G3, Pulse	99.82	99.84	99.87	99.85	0.32
DMS-BTT	C3 only	0.5 (paper)	G3, Pulse	99.82	99.84	99.87	99.85	0.32
FEVER	C1–C3 federated	0.1 % benign	G3, Pulse	99.77	99.70	99.91	99.81	0.79